

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch : Artificial Intelligence & Data Science /Computer Science and Engineering(Artificial Intelligence and Machine Learning)/Computer Science and Engineering (Artificial Intelligence)/Computer Science and Engineering(Data Science)

Semester :V

Subject Code & Name: BTAIHM503C Knowledge reasoning and AI ethics

Max Marks: 60

Date:05-01-24

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Discuss common challenges in knowledge representation?	L2/CO1	6
B) How can knowledge representation help an agent navigate through the Wumpus World?	L2/CO1	6
C) Explain the basic principles of propositional logic and its role in artificial intelligence.	L2/CO1	6
Q.2 Solve Any Two of the following.		12
A) Describe the resolution method in logic and its application in theorem proving.	L2/CO2	6
B) Explain the concept of natural deduction in logic.	L2/CO2	6
C) Describe the process of backward chaining in first-order logic.	L2/CO2	6
Q. 3 Solve Any Two of the following.		12
A) Discuss the challenges and strategies for making decisions when facing uncertainty.	L2/CO3	6
B) Explain Bayes' Rule and its application in probabilistic reasoning.	L2/CO3	6
C) Demonstrate How does the structure of a Bayesian network influence probabilistic reasoning?	L3/CO3	6
Q.4 Solve Any Two of the following.		12
A) Discuss the advantages and disadvantages of different AI implementation approaches.	L2/CO4	6
B) Examine the impact of AI on the legal system.	L4/CO4	6

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| C) | Discuss strategies for addressing ethical concerns related to data governance in AI. | L2/CO4 | 6 |
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Q. 5	Solve Any Two of the following.		12
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| A) | Discuss the ethical challenges associated with the deployment of autonomous systems. | L2/CO5 | 6 |
| B) | Discuss the role of regulatory bodies in shaping ethical guidelines for AI. | L2/CO5 | 6 |
| C) | Discuss ethical concerns and arguments against the use of AI in military applications. | L2/CO5 | 6 |

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